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Research Proposal

Reconstructed Jira histories for late sprint closure and unfinished committed scope

Background

Sprint Planning is a natural point for early warning because the team can still adjust scope before execution begins. Repository data may contain useful signals, but only if the features reflect what was known at that time.

Research aim

The study will examine whether Jira history can rank two repository-derived outcomes: late sprint closure and unfinished scope that was committed at sprint start. The work will also test whether simple scope measures explain most of the predictable signal.

Research questions

1. How much ranking signal is present for late closure and unfinished committed scope?
2. Does a transparent scope-only model perform as well as more flexible models?
3. How stable are the results across time and across projects?
4. How sensitive are the findings to missing historical provenance and leakage-prone evaluation choices?

Data

The planned dataset is TAWOS, a relational collection of public Jira projects. The working subset contains eight projects. Sprint, Issue, and Change_Log records are used so that selected fields can be reconstructed at the start of each sprint.

Method

Sprint membership and Story Points will be reconstructed from timestamped change history. Nonclosed sprints will be removed before labels are built. Models will include logistic regression, random forest, and XGBoost, alongside prevalence and scope-only baselines. Evaluation will use forward temporal splits and held-out-project transfer.

Evaluation

ROC-AUC will be the main ranking measure. F1, precision, recall, PR-AUC, and Brier score will be reported with their limitations. Project-level uncertainty, multiple temporal cut points, provenance sensitivity, feature permutation, and leakage demonstrations will be used to test the robustness of the main conclusions.

Expected contribution

The intended contribution is methodological as well as predictive: a reproducible example of how timing, scope provenance, prevalence, project heterogeneity, and leakage affect sprint-analytics results.

Limits

The study is retrospective and uses open-source repositories. It will not claim that a risk score improves Scrum decisions; that would require prospective validation with real teams.